

IITB ACADEMIC INSTI-ASSIST

A RAG-Powered AI Assistant for IIT Bombay Academics — Project Write-Up

1. Chosen Scope and Why

This project implements an Academic Assistant for IIT Bombay, covering the academic calendar, undergraduate curriculum structure, examination procedures, and disciplinary rules. This scope was chosen because academic policy documents are dense, rule-heavy, and highly specific — the exact kind of content where a general-purpose LLM tends to hedge, guess, or produce plausible-sounding but incorrect answers (e.g., exact submission deadlines, specific clause numbers, or procedural steps).

This made it a strong candidate for demonstrating the value of retrieval-augmented grounding: instead of relying on the model's training data, the assistant answers strictly from the actual institute documents, and explicitly declines to answer when the retrieved context doesn't support a confident response.

2. Data Sources Used

Five official IIT Bombay documents were collected and used as the knowledge base:

- **Academic Calendar (2026–27)** — semester dates, registration windows, and key academic deadlines.
- **UG Curriculum (Mechanical Engineering)** — course structure, sequencing, and graduation requirements.
- **Procedures Document (July 2015)** — formal academic and administrative procedures.
- **Punishments / Disciplinary Document (July 2015)** — institutional rules and consequences for academic misconduct.
- **UG Rulebook** — general undergraduate academic regulations; an exceptionally large document that served as a rigorous testbed for chunking validation and semantic depth.

These documents were chosen because they are official, publicly relevant to students, and collectively cover the core academic scope without overlapping so heavily that retrieval would become redundant.

3. Chunking Strategy and Why

Documents are first parsed into raw text using PyPDF2, then split using a custom sentence-aware chunking function. Text is split on sentence boundaries (rather than blindly by character count) to avoid cutting sentences mid-way, which would produce chunks that read as incoherent fragments and embed poorly. Chunks are built up to a target size of 500 characters, with a 50-character overlap carried from the end of one chunk into the start of the next.

The overlap was included specifically to reduce the risk of losing context at chunk boundaries — a rule or clause that begins near the end of one chunk and continues into the next is more likely to still be retrievable in full. A relatively small chunk size (500 characters) was chosen over larger chunks to keep retrieved

context focused and reduce the chance of irrelevant text diluting the prompt sent to the LLM, since these policy documents tend to pack a lot of distinct rules into a short span of text.

Each chunk is embedded using the `all-mpnet-base-v2` sentence-transformer model and stored in a persistent ChromaDB collection, along with metadata identifying its source document and chunk index — this metadata is what powers the source-attribution shown alongside every generated answer.

4. Known Limitations and What I'd Improve With More Time

Known limitations:

- PDF text extraction quality depends on source formatting — tables and multi-column layouts sometimes extract with garbled spacing or lost structure.
- Chunking is based on a fixed character size/overlap rather than document structure (headings, sections), so a chunk can occasionally split a rule awkwardly.
- Retrieval uses a fixed top-k of 2 with no re-ranking step, which may miss relevant chunks for more complex, multi-part questions.
- Only one embedding model was tested; no comparative benchmarking was done against alternatives.
- The knowledge base is limited to five documents and one scope (academics) — questions about hostel life, clubs, or other institute topics are correctly declined, but are simply out of scope for this version.

What I'd improve with more time:

- Add a method to maintain a conversational history with the LLM so each prompt would not be treated as a new conversation.
- Add a re-ranking step after initial retrieval to improve precision on ambiguous or multi-part queries.
- Chunk by document structure (headings/sections) instead of a fixed character count, to keep logically related rules together.
- Add a lightweight feedback mechanism so users can flag incorrect or unhelpful answers for future improvement.
- Expand the document set toward a broader "General Insti Assistant" scope, covering hostel life and student council activities.
- Benchmark multiple embedding models and top-k values to tune retrieval quality more rigorously.